

CS 255 Business Requirements Document Template

System Components and Design

Purpose

- DriverPass is a new company that aims to address the high failure rate (over 65%) of students taking their driving license exam at the DMV
- The client wants to provide comprehensive driver training through a combination of online practice exams, educational content, and on-the-road training
- The system must handle customer registration, appointment scheduling, online course delivery, practice test administration, and tracking of student progress
- The purpose is to create an integrated web-based platform that manages all aspects of driver training services from initial registration through completion

System Background

- *DriverPass identified a market gap where students fail driving tests because they only study previous tests without adequate practical training*
- *The system needs to provide three distinct service packages with varying levels of training (6, 8, or 12 hours of in-car training)*
- *Key components include: user authentication and management, online course platform with practice tests, reservation/scheduling system for driving lessons, customer information database, DMV integration for updated content, reporting and tracking functionality, and payment processing*
- *The system must accommodate multiple user types: customers, secretary/office staff, IT officer, and owner*
- *Must track which driver and vehicle are assigned to each lesson, along with scheduling details*
- *Integration with DMV is required to receive notifications about rule changes, policy updates, and new sample questions*

Objectives and Goals

- *Enable customers to access online practice exams and study materials that reflect current DMV requirements*
- *Allow customers to schedule, modify, and cancel on-the-road driving lesson appointments through multiple channels (online, phone, in-person)*
- *Provide a cloud-based platform accessible from any computer or mobile device with internet connection*
- *Track all user activities including reservations, modifications, and cancellations with clear audit trails*
- *Maintain current and accurate training content through automated DMV updates*

- *Support three customizable training packages that can be enabled/disabled as needed*
- *Deliver comprehensive reporting capabilities showing student progress, test scores, driver notes, and lesson history*
- *Ensure secure access with role-based permissions for different user types*
- *Facilitate efficient management of 10 vehicles and their assigned drivers*
- *Enable data access and report downloading for offline analysis*

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

- *The system must be web-based and run on cloud infrastructure to eliminate on-premise backup and security concerns*
- *Must be accessible from any computer or mobile device with internet connectivity*
- *Online data modifications must be synchronous to prevent data redundancy across servers*
- *System should provide real-time updates when customers make or modify reservations*
- *Must support concurrent access by multiple users (customers, staff, drivers) without performance degradation*
- *Reports must be downloadable in formats compatible with Excel for offline analysis*
- *System should provide instant notifications when DMV publishes content updates*
- *Interface must be responsive and load quickly on both desktop and mobile platforms*

Platform Constraints

- *System must run on web browsers across multiple platforms (Windows, Mac, mobile devices)*
- *Backend requires cloud hosting infrastructure to handle storage, backup, and security*
- *Must support integration with DMV systems for receiving automated updates*
- *Database backend required to store customer information, reservations, test results, and training content*
- *Must support secure payment processing for credit card transactions*
- *Platform should support standard web technologies accessible across modern browsers*
- *Mobile compatibility is essential for customers accessing the system on smartphones and tablets*

Accuracy and Precision

- *System must distinguish between different user roles (owner, IT officer, secretary, customer) with appropriate access levels*
- *User authentication must be case-sensitive and secure*
- *All reservation data must accurately track customer name, driver assignment, vehicle, date, time, and location*
- *System must maintain precise records of test scores, completion status, and timestamps*

- *Activity logs must capture exact details of who made changes and when (user ID, timestamp, action type)*
- *Payment information must be processed and stored with high accuracy and security*
- *Driver notes must be timestamped and associated with the correct lesson and student*
- *System should notify administrators of critical issues such as failed login attempts or system errors*
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Adaptability

- *IT officer must have full access to add, remove, modify, and disable user accounts without code changes*
- *Owner must be able to enable or disable training packages without developer intervention*
- *System architecture should support future addition or removal of modules by developers*
- *Must adapt to DMV content updates automatically without requiring system downtime*
- *User interface should be flexible enough to accommodate different screen sizes and devices*
- *System should handle platform and browser updates without requiring major redesign*
- *Database schema should be designed to accommodate future expansion of features*
- *IT officer needs administrative access to reset passwords, block accounts, and manage system settings*
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Security

- *Users must authenticate with username and password to access the system*
- *Passwords must meet security standards and be encrypted in the database*
- *Automatic password reset functionality required for users who forget credentials*
- *Role-based access control must restrict functionality based on user type*
- *IT officer must have ability to immediately block access for terminated employees*
- *Credit card information must be encrypted and stored securely following PCI compliance standards*
- *System must implement protection against brute force login attempts (account lockout after multiple failed attempts)*
- *All data transmission between client and server must use secure protocols (HTTPS/SSL)*
- *Activity tracking must log all modifications to provide accountability and security audit trails*
- *Cloud provider must handle backup and security infrastructure*
- *Session management must automatically log out inactive users after a specified period*

Functional Requirements

- *The system shall allow customers to create accounts by providing first name, last name, address, phone number, state, credit card information, and preferred pickup/drop-off location*
- *The system shall validate user credentials during login and grant access based on assigned role*

- *The system shall provide three training packages: Package One (6 hours driving), Package Two (8 hours driving plus in-person lesson), Package Three (12 hours driving, in-person lesson, and online course access)*
- *The system shall allow customers to register for one of the three available training packages*
- *The system shall enable customers to schedule two-hour driving lesson appointments online by selecting date, time, and preferences*
- *The system shall allow secretary to schedule appointments on behalf of customers via phone or in-person requests*
- *The system shall permit customers to modify or cancel existing appointments through their online account*
- *The system shall track and display which driver and vehicle are assigned to each appointment*
- *The system shall provide online practice tests that customers can take multiple times*
- *The system shall record test results including test name, time taken, score, and status (not taken, in progress, failed, passed)*
- *The system shall display student progress showing all tests with their completion status*
- *The system shall allow drivers to enter notes and comments after each lesson*
- *The system shall maintain a lesson history showing lesson time, start hour, end hour, and driver comments*
- *The system shall receive and process automated updates from DMV regarding rules, policies, and sample questions*
- *The system shall notify administrators when DMV updates are available*
- *The system shall allow owner to download reports and data for offline analysis in Excel-compatible formats*
- *The system shall generate activity reports showing who made, modified, or canceled reservations*
- *The system shall provide audit trail functionality tracking all user actions on records*
- *The system shall allow IT officer to reset user passwords and manage all user accounts*
- *The system shall enable IT officer to block user access when necessary*
- *The system shall allow owner to enable or disable training packages without code modifications*
- *The system shall process and securely store credit card payments including card number, expiration date, and security code*
- *The system shall display customer information including personal details, special needs, and photos (driver and student)*
- *The system shall provide contact forms for customers to communicate with DriverPass*
- *The system shall support both online and offline data access, with modifications only permitted when online*

User Interface

- *Primary users include customers (students seeking driver training), secretary (scheduling appointments and answering phones), IT officer (system administration and maintenance), and owner (business management and reporting)*
- *Interface must be accessible via web browsers on desktop computers, laptops, tablets, and mobile devices*
- *Cloud-based deployment eliminates need for local software installation*

- *Customer interface needs: registration forms for account creation, login page with authentication, dashboard showing personal information and test progress, online practice test interface, reservation system for scheduling/modifying appointments, contact forms, display of driver notes and lesson history*
- *Secretary interface needs: appointment scheduling system, customer information access, ability to create reservations on behalf of customers*
- *IT officer interface needs: administrative panel for user management, password reset functionality, account blocking capabilities, system configuration access*
- *Owner interface needs: comprehensive reporting dashboard, data export functionality, ability to enable/disable packages, access to all activity reports and audit trails*
- *Interface must display logo at top of pages per client specifications*
- *Main customer dashboard should include four sections: online test progress (showing test name, time taken, score, status), customer information (first name, last name, address, city, state, zip, phone, email), driver notes (lesson times and comments in table format), and special needs section with driver and student photos*
- *Input forms required for student information collection including all required fields*
- *Navigation must be intuitive allowing users to easily access their relevant functions*
- *Mobile interface should maintain functionality while being optimized for smaller screens*
- *System should provide clear feedback for user actions (confirmations, error messages, success notifications)*

Assumptions

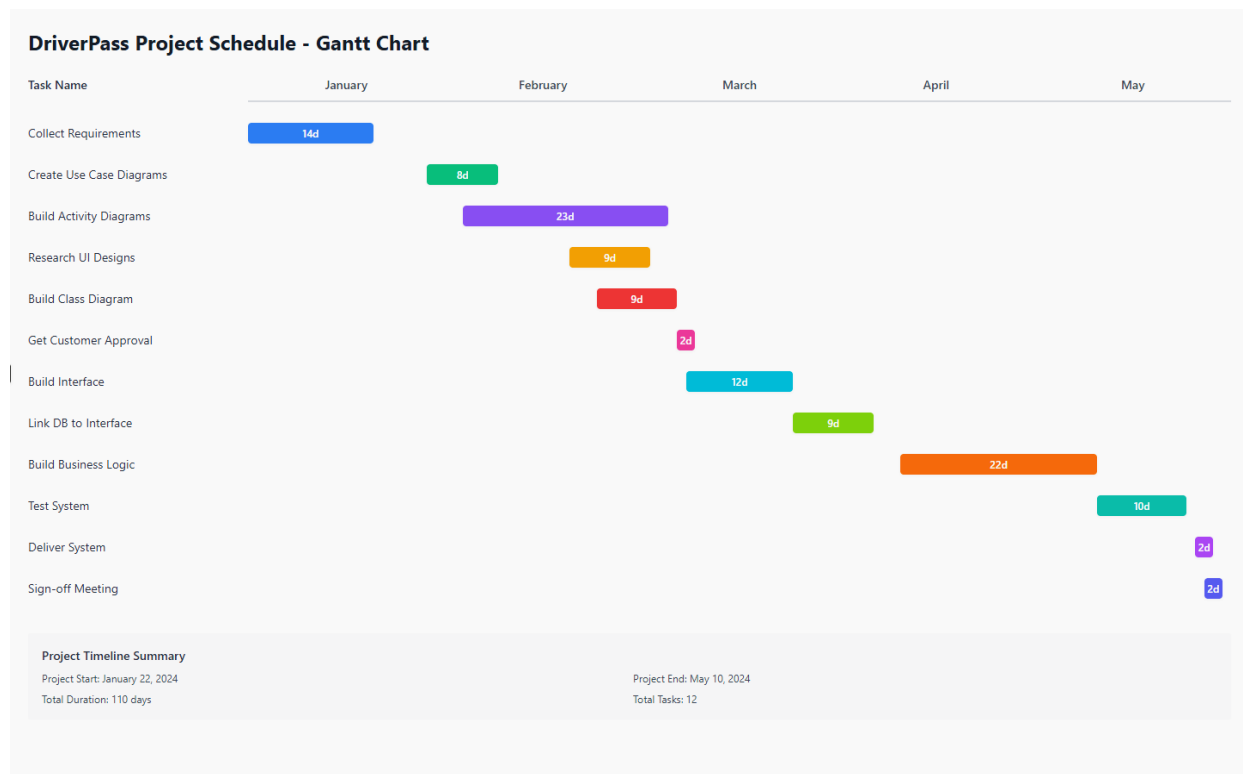
- *Customers have access to internet-enabled devices (computers, tablets, or smartphones) to use the online system*
- *Customers possess valid credit cards for payment processing*
- *DMV provides an API or automated method for sending content updates to third-party systems*
- *Cloud hosting provider offers adequate security, backup, and infrastructure to meet business needs*
- *Customers have basic computer literacy to navigate web-based interfaces*
- *Secretary and staff have computer access and training to use the system effectively*
- *The business has sufficient drivers and vehicles (10 cars) to accommodate scheduled appointments*
- *Appointment scheduling operates during defined business hours*
- *Customers provide accurate information during registration*
- *Internet connectivity is reliable for both business operations and customer access*
- *Standard two-hour lesson duration is appropriate for all training types*
- *Pickup and drop-off locations are within the service area of DriverPass*
- *Credit card processing fees are acceptable within the business model*
- *System will initially launch with the three defined packages before future customization*

Limitations

- *System modifications requiring new modules or significant functionality changes will need developer involvement and cannot be performed by non-technical staff*
- *Offline data access is read-only; modifications can only be made when online to prevent data synchronization issues*

- *Initial system will not include built-in package customization beyond enabling/disabling existing packages*
- *Two-hour lesson duration is fixed and cannot be adjusted without system modifications*
- *System depends on DMV providing timely and accurate updates; delays or errors from DMV will affect content currency*
- *Cloud provider limitations may affect storage capacity, performance, or uptime*
- *Project timeline constraints limit the scope to essential features, with advanced customization deferred to future releases*
- *Budget constraints may limit the extent of custom interface design or advanced features*
- *System requires stable internet connection for full functionality*
- *Initial implementation focuses on core business operations rather than future expansion features*
- *Mobile interface may have reduced functionality compared to desktop version depending on screen size constraints*
- *No ability for non-developers to add custom fields or modify database structure without technical assistance*

Gantt Chart



Please refer to the interactive Gantt Chart created above, which shows the complete project schedule from January 22 through May 10, including all 12 tasks with their dependencies and timelines. The chart displays:

- Collect Requirements (14 days)
- Create Use Case Diagrams (8 days)
- Build Activity Diagrams (23 days)
- Research UI Designs (9 days)
- Build Class Diagram (9 days)
- Get Customer Approval (2 days)
- Build Interface (12 days)
- Link DB to Interface (9 days)
- Build Business Logic (22 days)
- Test System (10 days)
- Deliver System (2 days)
- Sign-off Meeting (2 days)]